a)

$$\left(\begin{array}{ccc|c} 20 & 10 & 0 & 150 \\ 50 & 30 & 20 & 470 \\ 200 & 150 & 100 & 2150 \end{array}\right) \cdot 5 \quad \cdot 2$$

$$\left(\begin{array}{ccc|c} 10 & 5 & 0 & 75 \\ 10 & 6 & 4 & 94\ 19 \end{array}\right)$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 0 & 15 \\ 0 & 1 & 4 & 19 \\ 4 & 3 & 2 & 43 \end{array}\right) \cdot 2 \quad -$$

$$\begin{array}{ccc|c} 4 & 2 & 0 & 30 \\ & & & \\ 4 & 3 & 2 & 43\ 13 \end{array}$$

$$\left(\begin{array}{ccc|c} 10 & 5 & 0 & 75 \\ 0 & 1 & 4 & 19 \\ 0 & 1 & 2 & 13 \end{array}\right)$$

$0\ 0\ -2\ |\ -6$

$-2z = -6$

$z = 3$

$1y + 4z = 19$

$1y + 12 = 19$

$y = 7$

$10x + 5y = 75$

$10x + 35 = 75$

$10x = 40$

$x = 4$

4 Elüge Typ A

7 – – B

1 – – C

b)

$$\left(\begin{array}{ccc|c} 20 & 10 & 0 & 120 \\ 50 & 30 & 20 & 350 \\ 200 & 150 & 100 & 1600 \\ 4 & 3 & 2 & 32 \end{array}\right) \cdot 5 \quad \cdot 2$$

$$\begin{array}{ccc|c} 10 & 5 & 0 & 60 \\ 10 & 6 & 4 & 70\ 10 \end{array}$$

$$\left(\begin{array}{ccc|c} 2 & 1 & 0 & 12 \\ 0 & 1 & 4 & 10 \\ 4 & 3 & 2 & 32 \\ 0 & 1 & 2 & 8 \end{array}\right) \cdot 2$$

$$\begin{array}{ccc|c} 4 & 2 & 0 & 24 \\ & & & \\ 4 & 3 & 2 & 32\ 8 \end{array}$$

$0\ 0\ -2\ |\ -2$

$$\left(\begin{array}{ccc|c} 2 & 1 & 0 & 12 \\ 0 & 1 & 4 & 10 \\ 0 & 0 & -2 & -2 \end{array}\right)$$

$-2z = -2$

$z = 1$

$1y + 4z = 10$

$1y + 4 = 10$

$y = 6$

$2x + y = 12$

$2x + 6 = 12$

$2x = 6$

$x = 3$

3 Elüge Typ A

6 – – B

1 – – C